

Apurva

2020 grad

9:05 PM - 11:30 PM

Recording + Notes

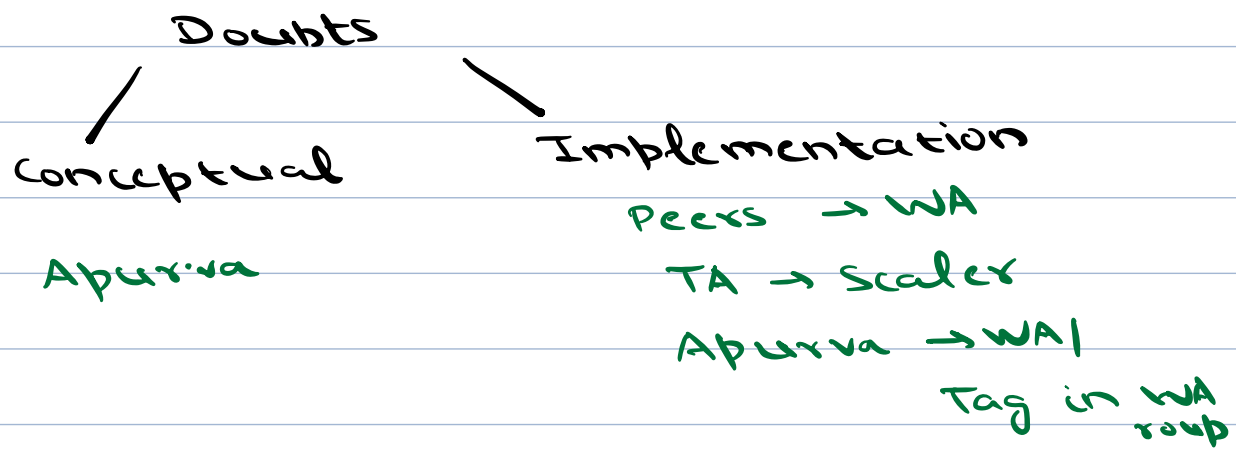
① PSP Problem solving Percentage ≥ 85	Assignment	Additional HW
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② Attendance ≥ 80
(Live + Recording)

ASK a ques $\rightarrow$ Public

Answer a ques $\rightarrow$ Private

Support @ [scaler.com](https://www.scaler.com)



- ① TC
- ② Arrays
- ③ TC, SC, Output
- ④ If sum carry forward
- ⑤ Ques
- ⑥ Sliding window, contri technique

Memory management

Sorting

2D matrices

Strings

Contest → 1.5 hours

Agenda -

1. Count of factors
2. Sum of N natural nos.
3. GP and sum of GP
4. Iterations
5. Comparing 2 algos
6. Big O - Time complexity

$$2 \sqrt[6]{\frac{1}{2}}$$



What is a factor?

$\Rightarrow$ i is a factor of N ?

When i divides N completely

if $(N \% i == 0)$

$\Rightarrow i$ is a factor of N

Factors of 8: $\textcircled{4}$ 1, 2, 4, 8

$N = 24$ $\textcircled{8}$ 1, 2, 3, 4, 6, 8, 12, 24

$N = 10$ $\textcircled{4}$ 1, 2, 5, 10

Minimum factor of any no.? 1

Maximum factor of any no.? N itself

$N \rightarrow$ all factors lie in range
1 to N

Pseudo code

function countFactors(N) $\leftarrow$

cnt = 0

for ($i \rightarrow 1$ to N) $\leftarrow$

// is i a factor of N

if $(N \% i == 0)$ $\leftarrow$

$N = 5$

$i =$
1
2
3
4
5

```

    cnt++
    return cnt

```

N iterations

Iterations $\rightarrow$ No. of times a loop runs

Code runs on servers

It takes some time to execute your code.

servers have the capability of executing 10^8 iterations in 1 sec

N	Iterations	Execution Time
10^8	10^8	1 s
10^9	10^9	10 s
10^{18}	10^{18}	10^{10} s
		317 years

①

$10^8 \rightarrow 1 \text{ s}$

$10^9 \rightarrow 10 \text{ s}$

10×10^8

$\boxed{3 \text{ choc}} \rightarrow \approx 1$

30 choc $\rightarrow \approx 10$

10 x

② Unitary Method

$$10^8 \text{ iterations} \rightarrow 1 \text{ sec}$$

$$1 \text{ iteration} \rightarrow \frac{1}{10^8} \text{ sec}$$

$$\frac{10^9}{10^8}$$

$$10^9 \text{ iterations} \rightarrow \frac{1}{10^8} \times 10^9 \text{ s} \\ = 10 \text{ s}$$

$$10^{18} \text{ iterations} = \frac{1}{10^8} \times 10^{18}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$= \frac{10^{18}}{10^8} \text{ s} \\ = 10^{10}$$

You? Children? 3rd? 4th? 5th...

Optimisation

$$\Rightarrow a * b = N \text{ (a and b are factors of N)}$$

$$\Rightarrow b = N/a \text{ (a and } N/a \text{ are factors of N)}$$

Factors come in pairs

$N = 24$

i	N/i	cnt
1	24	+2
2	12	+2
3	8	+2
4	6	+2
6	4	
8	3	
12	2	
24	1	

$$i \leq N/i$$

$$i * i \leq N$$

$$\Rightarrow i^2 \leq N$$

$$\Rightarrow i \leq \sqrt{N}$$

$$i_{\max} = \sqrt{N}$$

$$i_{\min} = 1$$

$$i \rightarrow [1, \sqrt{N}]$$

$N = 100$

i	N/i	cnt
1	100	+2
2	50	+2
4	25	+2
5	20	+2
10	10	+1
20	5	
25	4	
50	2	
100	1	

function countFactors(N) {

cnt = 0

for (i = 1; i ≤ √N; i++)

if (N % i == 0) {

cnt = cnt + 2

return cnt

√N

① Math.sqrt(N)

② pow(N, 0.5)

③ N^{0.5}

④ i * i ≤ N

function countFactors(N) <

```
    cnt = 0
    for (i = 1 ; i ≤ √N ; i++)
        if (N % i == 0) <
            if (i == N/i) <
                cnt = cnt + 1
            else <
                cnt = cnt + 2
    return cnt
```

$\sqrt{N}$ iterations

N
 10^{18}

Iterations

$\sqrt{10^{18}}$

$= 10^{18/2} = 10^9$

Execution Time

10 s

317 yrs $\rightarrow$ 10 s

Q. Given N , check if it is prime or not?

10, 11, 23, 2, 29, 27, 31 cnt = 4
✗ ✓ ✓ ✓ ✗ ✗ ✓

A prime no. has only 2 factors

↓
1 and no. itself

★ 1 is neither prime nor composite

bool checkPrime(N) {

if (countFactors(N) == 2)

return true

else

return false



10:40

Gauss → 4th class

$$S = 1 + 2 + 3 + 4 + 5 + 6 + \dots + 99 + 100$$

$$S = 100 + 99 + 98 + 97 + \dots + 2 + 1$$

$$2S = 101 + 101 + 101 + 101 + \dots + 101 + 101$$

$$2S = 101 \times 100$$

$$S = \frac{101 \times 100}{2} = \frac{101 \times \overset{50}{\cancel{100}}}{2} = 5050$$

Generalize this for first N natural nos.

$$S = 1 + 2 + 3 + \dots + (N-1) + N$$

$$S = N + (N-1) + (N-2) + \dots + 2 + 1$$

$$2S = (N+1) + (N+1) + (N+1) + \dots + (N+1)$$

$$2S = (N+1) \times N$$

$$S = \frac{N(N+1)}{2}$$

$[\rightarrow$ inclusive , $(\rightarrow$ exclusive

① $[a, b] \rightarrow$ Range means a and b both are included

② $(a, b) \rightarrow$ Range means a and b both are excluded

$[2, 5] \rightarrow 2, 3, 4, 5$

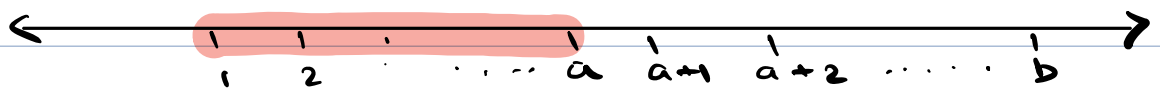
$(2, 5) \rightarrow 3, 4$

$[2, 5) \rightarrow 2, 3, 4$

Quiz: $[3, 10] : 3, 4, 5, 6, 7, 8, 9, 10$

ans $\rightarrow 8$

Quiz: $[a \quad b]$ $b - a + 1$ $(a \quad b)$
 $[a \quad b)$



Iterations: No. of times a loop runs

Ex: for ($i=1$; $i \leq N$; $i++$) $\{$

if ($i == N$) $\{$
break
 $\}$
 $\}$

$i=1$ ✓
 $i=2$ ✓
 $i=3$ ✓

$i: [1 \quad N] = N - 1 + 1 = N$ $O(N)$
iterations

Ex: for ($i=0$; $i \leq 100$; $i++$) <

| $s = s + i + i^2$
|>

$$[a \ b] \\ = b - a + 1$$

$$i: [0, 100] \rightarrow 100 - 0 + 1 \\ = 101 \text{ iterations}$$

$$101 \times N^0 \rightarrow N^0 \quad O(1)$$

Ex: for ($i=1$; $i \leq N$; $i++$) <

|>

$$i: [1 \ N] \rightarrow N \text{ iterations}$$

for ($j=1$; $j \leq M$; $j++$) <

|>

$$j: [1 \ M] \rightarrow M \text{ iterations}$$

$$\text{Total} = N + M \text{ iterations}$$

$$O(N + M)$$

Geometric Progression (Series)

$$\begin{array}{ccccccc} 5 & \xrightarrow{\times 2} & 10 & \xrightarrow{\times 2} & 20 & \xrightarrow{\times 2} & 40 & \xrightarrow{\times 2} & 80 & \xrightarrow{\times 2} & 160 \\ \underbrace{10/5=2} & & \underbrace{20/10=2} & & \underbrace{40/20=2} & & \underbrace{80/40=2} & & & & \end{array}$$

In a GP, common ratio is same

$$a=5 \quad r=2 \quad n=6$$

r = common ratio

a = 1st term of GP

n = no. of terms

Generic GP

$$a \quad ar \quad ar^2 \quad ar^3 \quad \dots \quad ar^{N-1}$$

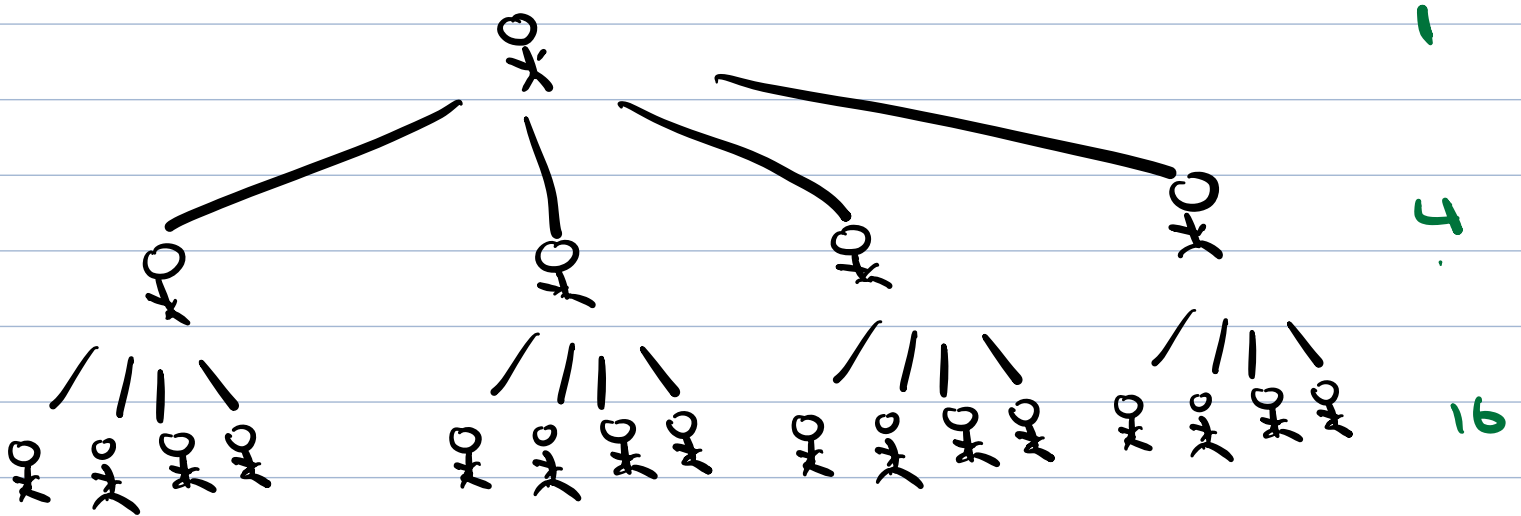
$N^{\text{th}} \text{ term}$

sum of first n terms of GP

$$= \frac{a(r^n - 1)}{r - 1}$$

$$r \neq 1$$

If $r = 1$ sum of n terms $= a \times n$



$$1, 4, 16, 64, \dots$$

$$a = 1$$

$$r = 4$$

$$N = ?$$

$$N^{\text{th}} \text{ term} = ar^{N-1}$$

Important

No. of people at 8th level =

$$1 \times 4^{8-1} = 4^7$$
$$= 16384$$

Important

Sum of people till 3rd level =

$$\frac{a(x^N - 1)}{x - 1}$$

$$\frac{1(4^3 - 1)}{4 - 1} = \frac{63}{3} = 21$$

$$a \in S = \langle 3, 2, 6, 8, 17 \rangle \rightarrow \langle 1, 2, 3, 6, 8 \rangle$$

Tharun
Algo1

15 s



windows xp



7 s

MBP M3

C++



7 s

Hot country



5 s

MT Everest

Sandhya
Algo2

10 s



MBP M3



10 s

Python



5 s

C++

MT Everest



5 s

Execution time is not a good parameter
for comparing algo

Execution time depends on OS / Lang /
Environment

To compare algo → No. of iterations

Compare two algorithms using iterations

↓
in terms of N

Algo1 (Muthu)

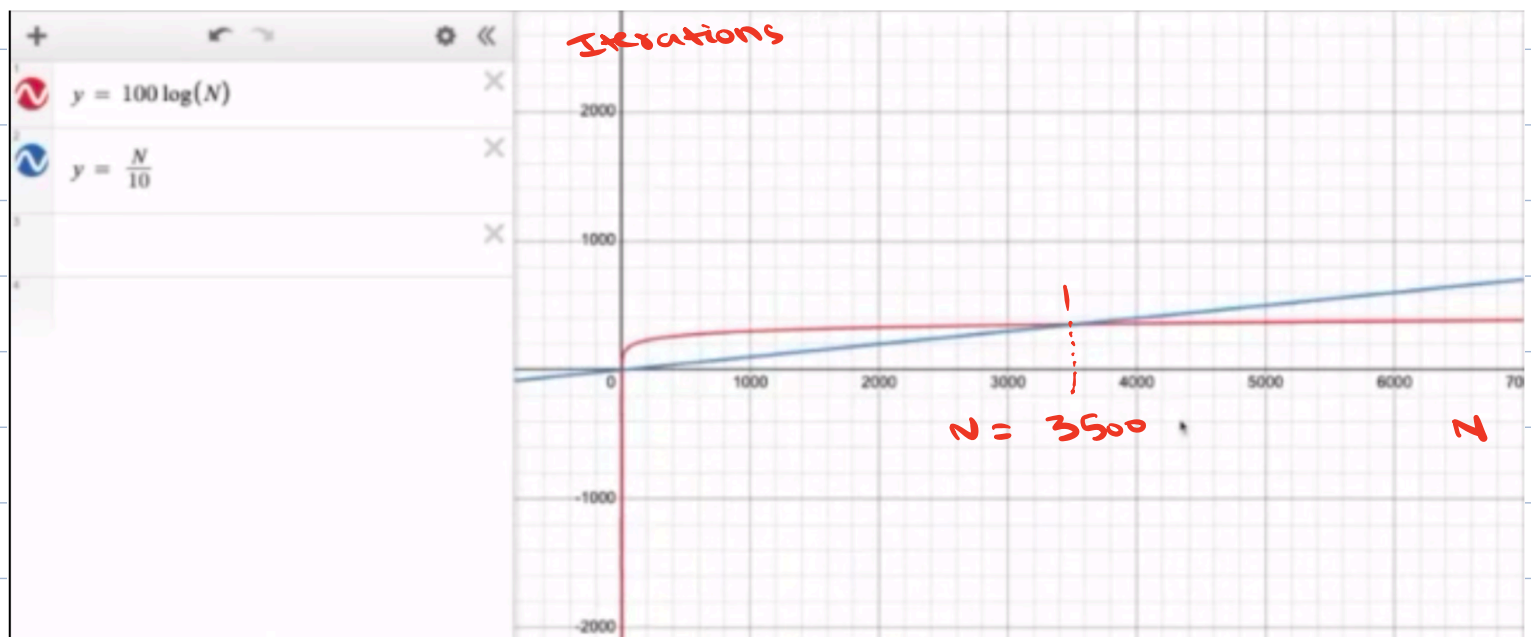
$$100 \log_2 N$$

$$\rightarrow O(\log_2 N)$$

Algo2 (Niraj)

$$N/10$$

$$\rightarrow O(N)$$



For $N < 3500$ Niraj < Muthu Niraj
 $N > 3500$ Muthu < Niraj Muthu

Lesser Iterations $\rightarrow$ Time

Data will increase over time
↓

Pick algo which works better
for larger input (Muthu)

Asymptotic analysis of algo



analyse performance of algo on large input



Big O Notation $O() \rightarrow TC$

Steps for Big O

1. Calculate no. of iterations

2. Ignore lower order terms

OR

Take highest order term

order
↓
power
highest
power of
 n

3. Ignore constant coefficient

$$f(n) = 4n^2 + 3n + 1$$

$$a^0 = 1$$

$$\downarrow$$
$$4n^2 + 3n^1 + 1n^0$$

$$\downarrow$$
$$4n^2 \rightarrow n^2$$

$$O(n^2)$$

Choose algo which has lower value of
Big O $\rightarrow TC$

Comparison order:

$$1 < \log N < \sqrt{N} < N < N \log N < N\sqrt{N} < N^2 < \dots \\ \dots N^3 < 2^N < N! < N^N$$

$$f(N) = 4N^2 + 3N + 6\sqrt{N} + 9\log_2 N + 10N^0$$

$\downarrow$
 $4N^2 \rightarrow N^2$ $O(N^2)$

$$f(N) = 4N + 3N \log_2 N + 1$$

$\downarrow$
 $3N \log_2 N \rightarrow N \log N$
 $O(N \log N)$

$$f(N) = 4N \log N + 3N\sqrt{N} + 10^6 N^0$$

$\downarrow$
 $3N\sqrt{N} \rightarrow N\sqrt{N}$
 $O(N\sqrt{N})$

For constant iterations $\rightarrow O(1)$

① $\log$

② $\log n$

for ($i=1$; $i \leq n$; $i=i*2$) <

1 —
↓

$n \rightarrow$